「Background.

Let f be an analytic on $B_r(z_0)\setminus\{z_0\}$. Then, there is the Laurent's Expansion:

$$f(z)=\sum_{n=-\infty}^{\infty} a_n\cdot(z-z_0)^n \quad (0<|z-z_0|<r)$$

where $a_n=\dfrac{1}{2\pi i}\cdot\displaystyle\int_C \frac{f(z)}{(z-z_0)^{n+1}}\,dz$ $(n\in\mathbb{Z})$, C is any closed contour around z_0.

So that $a_{-1}=\dfrac{1}{2\pi i}\cdot\displaystyle\int_C f(z)\,dz \overset{\text{def}}{=} \mathrm{Res}(f;z_0)$. Therefore, if we can find the Laurent's expansion, then we can evaluate integral.

「Thm. Suppose that f is analytic inside and on a simple closed contour C, except for isolated singularities at $z_1,\cdots,z_n$ inside C. Then

$$\int_C f(z)\,dz = 2\pi i\sum_{k=1}^{n}\mathrm{Res}(f,z_k).$$

There are two useful Lemmas.

「Lemma. If f has a pole of order m at z_0, then

$$\mathrm{Res}(f,z_0)=\frac{1}{(m-1)!}\cdot\lim_{z\to z_0}\left(\frac{d}{dz}\right)^{k-1}(z-z_0)^k f(z).$$

Proof. Observe that $f(z)=\dfrac{a_{-m}}{(z-z_0)^m}+\cdots+\dfrac{a_{-1}}{z-z_0}+a_0+a_1(z-z_0)+\cdots$ $(a_{-m}\neq 0)$

So that $(z-z_0)^m\cdot f(z)=a_{-m}+a_{-m+1}(z-z_0)+\cdots+a_{-1}(z-z_0)^{m-1}+g(z)\cdot(z-z_0)^m$,
where g is analytic at z_0.
Now, differentiating $m-1$ times, and evaluating z_0. ☒

「Lemma. Let f and g be analytic at z_0.
If g has a simple zero at z_0 and $f(z_0)\neq 0$, then

$$\mathrm{Res}(f/g,z_0)=\frac{f(z_0)}{g'(z_0)}$$

Proof. Since g has simple zero at z_0, $1/g$ has a simple pole at z_0. ☒

Examples.

$$\int_{|z|=\pi} \frac{e^z-1}{1-\cos z}\,dz = 4\pi i.$$

Since $\cos z = \cos\left(\frac{z}{2}+\frac{z}{2}\right) = \cos^2\frac{z}{2} - \sin^2\frac{z}{2} = 1-2\cdot\sin^2\frac{z}{2}$, So $\frac{e^z-1}{1-\cos z} = \frac{e^z-1}{2\sin^2\left(\frac{z}{2}\right)}$

And, e^z-1 has only one zero at $z=0$, and inside $|z|=\pi$, $\sin^2\left(\frac{z}{2}\right)=0 \Leftrightarrow z=2m\pi$ $(m\in\mathbb{Z})$

So it has only one isolated singularity inside $|z|=\pi$, at $z_0=0$.

Moreover, the zero of e^z-1 is order of 1, and zero of $\sin^2(z)$ has order 2,

So the given function has a simple pole at 0.

Now, $\lim_{z\to 0} z\cdot\frac{e^z-1}{2\cdot\sin^2\left(\frac{z}{2}\right)} = 2$. Therefore, $\operatorname{Res}(f,0)=2$.

Consequently, the residue theorem gives the result.

$$\int_{|z|=\pi} \frac{\sin z}{1-\cos z}\,dz$$

Since $1-\cos z = 2\cdot\sin^2\left(\frac{z}{2}\right)$ and $\sin z = 2\cdot\sin\left(\frac{z}{2}\right)\cdot\cos\left(\frac{z}{2}\right)$,

$$\frac{\sin z}{1-\cos z} = \frac{\cos\frac{z}{2}}{\sin\frac{z}{2}}.$$

Example.
Evaluate the integral $\displaystyle\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}$.

Construct a Curve C as piece-wise: given $\varepsilon, R > 0$,

$C_1 : z_1(t) = \varepsilon \cdot e^{-it} \ (-\pi \le t \le 0)$
$C_2 : z_2(t) = t \quad (\varepsilon \le t \le R)$
$C_3 : z_3(t) = R \cdot e^{it} \ (0 \le t \le \pi)$
$C_4 : z_4(t) = t \quad (-R \le t \le -\varepsilon)$.

$\sin z$
$\displaystyle = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{z^{2n+1}}{(2n+1)!} = \frac{e^{iz} - e^{-iz}}{2i}$.

And $C = C_1 \cdot C_2 \cdot C_3 \cdot C_4$. (The composite.)
Now, the function $f(z) = \dfrac{e^{iz}}{z}$ is analytic inside and on C. So,

$$\int_C f(z)\,dz = \sum_{i=1}^{4} \int_{C_i} f(z)\,dz$$

$$= \int_{-R}^{-\varepsilon} \frac{e^{it}}{t}\,dt + \int_{\varepsilon}^{R} \frac{e^{it}}{t}\,dt + \int_{-\pi}^{0} \frac{e^{i\cdot\varepsilon\cdot e^{-it}}}{\varepsilon e^{-it}} \cdot \varepsilon \cdot (-i) \cdot e^{-it}\,dt + \int_0^{\pi} \frac{e^{i\cdot R\cdot e^{it}}}{R\cdot e^{it}} \cdot R \cdot i \cdot e^{it}\,dt$$

$$= \int_{\varepsilon}^{R} \frac{e^{ix} - e^{-ix}}{x}\,dx - i \cdot \int_{-\pi}^{0} e^{iR\cdot e^{it}}\,dt + i \cdot \int_0^{\pi} e^{i\varepsilon\cdot e^{it}}\,dt.$$

$= 0$ (Cauchy's Theorem).

Meanwhile, $\displaystyle \left| i \cdot \int_0^{\pi} e^{i\cdot R\cdot e^{i\theta}}\,d\theta \right| \le \int_0^{\pi} \left| e^{i\cdot R\cdot e^{i\theta}} \right| d\theta \le \int_0^{\pi} e^{-R\cdot\sin\theta}\,d\theta = 2 \cdot \int_0^{\frac{\pi}{2}} e^{-R\sin\theta}\,d\theta$.

$\displaystyle \le 2 \cdot \int_0^{\frac{\pi}{2}} e^{-\frac{2R\theta}{\pi}}\,d\theta = 2 \cdot \left(-\frac{\pi}{2R}\right) \cdot \left(e^{-R} - 1\right)$

$0 \le t \le \frac{\pi}{2} \Rightarrow \sin t \ge \frac{2}{\pi} t$.

$\displaystyle = \frac{1 - e^{-R}}{R} \cdot \pi \to 0$ as $R \to \infty$.

And, $\displaystyle e^{i\varepsilon e^{it}} = \sum_{n=0}^{\infty} \frac{1}{n!} \cdot (i\varepsilon e^{it})^n$ follows $\displaystyle \left| e^{i\varepsilon e^{it}} - 1 \right| \le \sum_{n=1}^{\infty} \frac{1}{n!} \cdot \varepsilon^n = e^{\varepsilon} - 1$, thus

$$\int_{-\pi}^{0} e^{i\varepsilon\cdot e^{it}}\,dt = \int_{-\pi}^{0} e^{i\varepsilon e^{it}} - 1\,dt + \int_{-\pi}^{0} 1\,dt \to \pi \text{ as } \varepsilon \to 0.$$

Consequently, letting $R \to \infty$ and $\varepsilon \to 0$, we obtain

$$\int_0^\infty \frac{e^{ix} - e^{-ix}}{x}\,dx = i\pi.$$

Since $\sin x = (e^{ix} - e^{-ix})/2i$, we obtain

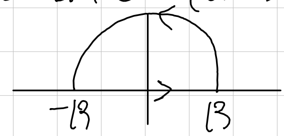
$$\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}.$$

「Evaluate $I = \int_0^\infty \frac{x \cdot \sin ax}{x^2+m^2}\,dx$ $\quad (a, m > 0)$.

Since the function in integral is even, $I = \frac{1}{2} \cdot \int_{-\infty}^{\infty} \frac{x \sin ax}{x^2+m^2}\,dx$.

Let C be the contour, Composite $t \mapsto t$ $(-R \le t \le R)$ and $t \mapsto Re^{it}$ $(0 \le t \le \pi)$,
Then C is a closed simple curve. Meanwhile, the function

$$f(z) = \frac{z e^{iaz}}{z^2+m^2}$$



The $z^2+m^2 = (z+im)(z-im)$ has only one zero in the upper half-plane, im,
so f has only one single pole inside C, for $R > m$. Moreover,

$$\operatorname{Res}(f, im) = \lim_{z \to im} (z-im) \cdot \frac{z e^{iaz}}{z^2+m^2} = \frac{im \cdot e^{-am}}{2im} = \frac{1}{2} \cdot e^{-am}.$$

By the Residue Theorem, for Γ_R, the semicircular part of C,

$$\int_C f\,dz = \int_{-R}^{R} f(x)\,dx + \int_{\Gamma_R} f(z)\,dz = \pi i \cdot e^{-am} \quad \cdots (*)$$

And since

$$\left| \int_{\Gamma_R} f(z)\,dz \right| = \left| \int_0^\pi \frac{Re^{it} \cdot e^{ia \cdot Re^{it}}}{R^2 \cdot e^{2it} + m^2} Rie^{it}\,dt \right| \qquad R^2+m^2 \ge R^2-m^2$$

$$\le \frac{R^2}{R^2-m^2} \cdot \int_0^\pi e^{-aR \cdot \sin t}\,dt$$

$$= \frac{R^2}{R^2-m^2} \cdot 2 \cdot \int_0^{\frac{\pi}{2}} e^{-aR \sin t}\,dt \qquad \frac{2}{\pi}t \le \sin t, \text{ in } 0 \le t \le \frac{\pi}{2}.$$

$$\Downarrow$$

$$\le \frac{2R^2}{R^2-m^2} \cdot \int_0^{\frac{\pi}{2}} e^{-aR \cdot \frac{2}{\pi} t}\,dt$$

$$= \frac{2R^2}{R^2-m^2} \cdot \frac{\pi}{-2aR} \cdot \left(e^{-aR} - 1\right) \to 0 \text{ as } R \to \infty.$$

So, letting $R \to \infty$ in $(*)$, we obtain

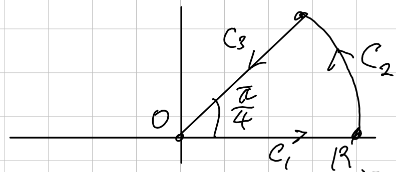
$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{\infty} \frac{x}{x^2+m^2} \left(\cos ax + i \cdot \sin ax\right) = \pi e^{-am} i.$$

Compare the imaginary part, we obtain that

$$I = \frac{1}{2} \cdot \pi \cdot e^{-am}.$$

「Evaluate

$$\int_0^\infty \sin x^2\,dx = \int_0^\infty \cos x^2\,dx = \frac{\sqrt{\pi}}{2\sqrt{2}}.$$



Let C be a curve, composite of

$C_1: t \mapsto t \ (0 \le t \le R)$, $C_2: t \mapsto Re^{it} \ (0 \le t \le \frac{\pi}{4})$, $C_3: t \mapsto -t \cdot e^{\frac{i\pi}{4}} \ (-R \le t \le 0)$

Since e^{iz^2} is entire, being composition of entire, we can apply the Cauchy's Thm

$$0 = \int_C e^{iz^2}\,dz = \int_0^R e^{it^2}\,dt + \int_0^{\frac{\pi}{4}} e^{i(Re^{it})^2} \cdot Rie^{it}\,dt + \int_{-R}^0 -e^{i \cdot t^2 \cdot e^{i\pi/2}} \cdot e^{\frac{i\pi}{4}}\,dt$$

Since $e^{i\pi/2} = i$,

$$\int_0^R e^{it^2}\,dt + \int_0^{\frac{\pi}{4}} e^{i(Re^{it})^2} \cdot Rie^{it}\,dt = \int_{-R}^0 e^{-t^2}\,dt \cdot e^{\frac{i\pi}{4}} = e^{\frac{i\pi}{4}} \int_0^R e^{-t^2}\,dt.$$

Meanwhile, observe that

$$\begin{aligned}\left| \int_0^{\frac{\pi}{4}} e^{iR^2 e^{2it}} \cdot Rie^{it}\,dt \right| &\le R \cdot \int_0^{\frac{\pi}{4}} \left| e^{iR^2 e^{2it}} \right| dt \\ &= R \cdot \int_0^{\frac{\pi}{4}} e^{-R^2 \cdot \sin 2t}\,dt \\ &= \frac{R}{2} \cdot \int_0^{\frac{\pi}{2}} e^{-R^2 \cdot \sin t}\,dt \\ &\le \frac{R}{2} \cdot \int_0^{\frac{\pi}{2}} e^{-R^2 \cdot \frac{2t}{\pi}}\,dt \\ &= \frac{R\pi}{4R^2}\left(1 - e^{-R^2}\right) \to 0 \text{ as } R \to \infty,\end{aligned}$$

Therefore, letting $R \to \infty$, we obtain

$$\begin{aligned}\int_0^\infty e^{ix^2}\,dx &= e^{\frac{i\pi}{4}} \cdot \int_0^\infty e^{-x^2}\,dx = \left(\frac{1}{\sqrt{2}} + i \cdot \frac{1}{\sqrt{2}}\right) \cdot \frac{\sqrt{\pi}}{2} \\ &= \int_0^\infty \cos x^2\,dx + i \cdot \int_0^\infty \sin x^2\,dx\end{aligned}$$

So, comparing the imaginary part, we obtain that

$$\int_0^\infty \sin x^2\,dx = \frac{\sqrt{\pi}}{2\sqrt{2}}.$$